

How to Analyse Big Data

Using the ELK Stack

This article introduces users to the ELK stack. Apart from discussing how it helps in better data analytics, there are detailed explanations about the internal workings of Elasticsearch, as well as insights into how Logstash and Kibana complement Elasticsearch's capabilities.



Big Data refers to large data sets that combine structured, semi-structured and non-structured data from various sources. Big Data is not just big in size but can also provide various insights about running and improving businesses when it is analysed. It can give insights on user behavioural patterns, changing trends, a system's health or any anomalies. The analysis helps improve customer integration, detect abnormalities and mitigate fraud.

Increased Internet penetration and a mobile in every hand are generating loads of user data for companies. This data is

of high velocity, volume and variety. A lot of data is being generated from mobiles, media devices, traffic surveillance systems, messages, etc. This data needs to be analysed in real-time to provide the user with consumable insights.

Imagine an e-commerce company that aims to provide high-quality customer service. When users search for a product on the website, they want the best-suited results to be shown. For a better user experience, various other factors like user age, brand, budget and location are added to the user search.

Relational databases have predefined schema to store structured

data, whereas data being generated today can be non-structured. The real power of data can be harnessed only when we are able to aggregate the data collected from various sources. Data can be structured (has a defined schema) or unstructured (has no defined schema). Examples like user number, location and email can be stored in an SQL database as structured. User information, such as payment mode, wish lists and previous orders, is stored in an unstructured format. To provide the best user experience, users should be shown recommendations and search results based on an analysis of the data